Decoupling Audience Engagement from critical praise

An Exploratory Analysis of IMDb Data(2010-2019

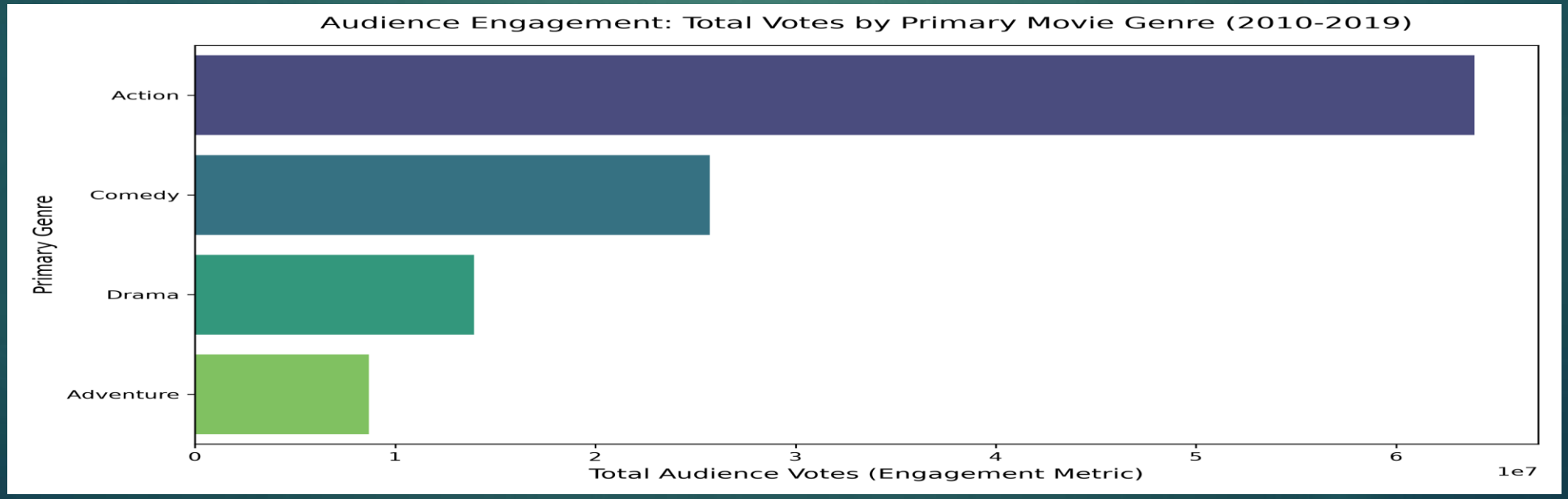
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Summary of Initial Data Exploration Findings

- **Volume vs. Engagement: The** genres produced in the highest volumes (such as Documentaries and Independent Dramas) do not drive the highest audience engagement.
- **The Blockbuster** Bias: Action, Adventure, and Sci-Fi primary genres overwhelmingly dominate total audience voting counts (Num votes), indicating dense viewer engagement despite mid-tier critical scores.
- **The Critical Da**rlings: Niche genres and documentaries consistently net higher average ratings (average rating) but suffer from low voter turnout, indicating small but highly dedicated audiences.
- **Temporal Boundaries**: Outlier records listing release years past 2019 were successfully identified and pruned from the workspace to preserve historical sample integrity.

Business Question Formulation (With Supporting Visualization)

- Substantial Business Question: "When greenlighting films, should a production studio prioritize broad audience engagement over critical acclaim? Specifically, how does a movie's primary genre choice affect its audience voting volume (Num votes) versus its average rating (average rating), and are there specific directors who consistently buck this trend by achieving both high volume and high critical praise?“
- Strategic Relevancy: Studios can use this dynamic to evaluate whether an upcoming project is built for mass market saturation (high engagement) or prestige/award positioning (high score, lower footprint).



Data Cleaning Assessment & Null Value Strategy

- **Multi-Genre Aggregation: The genres column frequently concatenates up to three distinct genres as a comma-separated** string (e.g., "Action,Adventure,Sci-Fi"). Deeper analysis requires a crosswalk extraction to isolate primary genres or pivot the entries out cleanly.
- **Null Value Exclusions**: Key structural records containing missing values in average rating, numvotes, or genres were cleanly filtered out of active query computations rather than filled with arbitrary placeholding statistics.
- **Personnel Entity Resolution**: The tables directors and writers isolate personnel via text codes (person_id). To make business reports human-readable, an operational JOIN structure against the persons table is required to pull authentic names.
- **Name Variant Standardization**: The movie akas table reveals that titles vary significantly by region. Analytical pipelines must lock down a normalized baseline region (such as standard domestic titles) to prevent duplicate metric counting.